

ABC MANUFACTURING CORPORATION

Environmental Health & Safety Department

STANDARD OPERATING PROCEDURE

Hazard Communication (HazCom)

Document No. SOP-EHS-002

Field	Detail	Field	Detail
Document Number	SOP-EHS-002	Revision Number	02
Effective Date	15 Jan 2026	Review Date	15 Jan 2028
Document Owner	Manager, EHS	Version	2.0
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A. Document Control

A.1 Revision History

Rev	Date	Description of Change	Author	Approver
00	03 Apr 2019	Initial issue of the written Hazard Communication program.	J. Alvarez	M. Okafor
01	11 Sep 2022	Updated to GHS label element requirements; added secondary-container labeling rules and pictogram reference.	R. Nguyen	M. Okafor
02	15 Jan 2026	Biennial review. Added risk matrix, non-routine task procedure, contractor coordination, and SDS-access emergency provisions.	J. Alvarez	M. Okafor

A.2 Distribution List

Copy No.	Holder / Role	Location	Format
Master	Document Owner — Manager, EHS	DMS (electronic master)	Electronic
01	Production Floor Supervisors	Line 1–4 supervisor stations	Controlled print
02	Chemical Store / Receiving Lead	Chemical storage building	Controlled print
03	Training Coordinator	Learning Management System	Electronic
04	Purchasing / Procurement	Procurement office	Electronic
05	Contractor Coordination Desk	Main gate / contractor sign-in	Controlled print

A.3 Controlled Copy Statement

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1. Purpose

This Standard Operating Procedure establishes the written Hazard Communication (HazCom) program for ABC Manufacturing Corporation. Its purpose is to ensure that the hazards of all chemicals produced or present in the workplace are evaluated and that information concerning those hazards is transmitted to affected employees and contractors through container labeling, Safety Data Sheets (SDS), and training, consistent with the Globally Harmonized System (GHS) of classification and labeling.

- Ensure every employee has ready access to hazard information for the chemicals they may be exposed to.
- Define requirements for chemical inventory, SDS management, workplace labeling, and employee training.
- Establish accountability for maintaining the written program and keeping it current.
- Support the site ISO 45001 and ISO 14001 management-system objectives.

2. Scope

This procedure applies to all ABC Manufacturing employees, contractors, and temporary workers who are or may be exposed to hazardous chemicals under normal operating conditions or in foreseeable emergencies, at all site locations.

2.1 Inclusions

- The written HazCom program, chemical inventory, SDS acquisition and access, and container labeling (primary and secondary/workplace containers).
- Employee information and training, non-routine task hazard communication, and multi-employer (contractor) coordination.

2.2 Exclusions

- Articles, food, and consumer products used in the same manner and duration as normal consumer use, as excepted under the standard.
- Hazardous waste and its labeling, which are governed by the site waste-management procedure and applicable EPA requirements.

3. References

This procedure is based on publicly available regulations and consensus standards. The most stringent applicable requirement governs.

Reference	Title / Subject
29 CFR 1910.1200	Hazard Communication (HazCom / GHS)
29 CFR 1910.1200 App. C	Allocation of label elements
29 CFR 1910.1200 App. D	Safety Data Sheets (16-section format)
GHS (UN)	Globally Harmonized System of Classification and Labelling of Chemicals
NFPA 704	Identification of the Hazards of Materials for Emergency Response
ANSI Z535.4	Product Safety Signs and Labels
ISO 45001:2018	Occupational health and safety management systems
ISO 14001:2015	Environmental management systems
Internal	SOP-EHS-001 Chemical Handling & Storage; SOP-EHS-003 Spill Response; SOP-EHS-013 Incident Reporting

4. Definitions

Term	Definition
Hazardous Chemical	Any chemical classified as a physical or health hazard, a simple asphyxiant, combustible dust, pyrophoric gas, or hazard not otherwise classified.
Safety Data Sheet (SDS)	The standardized 16-section document describing a chemical's identity, hazards, handling, and emergency measures.
GHS	Globally Harmonized System — the international framework for hazard classification and standardized label elements.
Pictogram	A GHS symbol on a white background within a red diamond border, conveying a hazard class.
Signal Word	"Danger" (more severe) or "Warning" (less severe), indicating the relative level of hazard on a label.
Hazard Statement	Standardized phrase describing the nature and degree of a hazard (e.g., "Causes serious eye damage").
Precautionary Statement	Standardized phrase describing recommended measures to minimize or prevent adverse effects.
Primary Container	The manufacturer's original labeled container as received.
Secondary / Workplace Container	A container into which a chemical is transferred from the primary container for use within the site.

5. Responsibilities

Role	Responsibility
EHS Manager (Document Owner)	Maintains the written program, ensures SDS system and training are functioning, conducts periodic program audits.
Training Coordinator	Delivers and documents HazCom training on hire, on new-hazard introduction, and on program change.
Area Supervisor	Ensures workplace containers are labeled, SDS access is available, and only trained employees handle chemicals.
Chemical Store / Receiving Lead	Maintains the chemical inventory and confirms an SDS is on file before a new chemical enters service.
Procurement	Requires suppliers to provide compliant labels and current SDS with each shipment.
Employee	Reads labels and SDS, labels secondary containers, and reports missing labels/SDS or new hazards.
Contractor Coordinator	Exchanges hazard information with on-site contractors (chemicals brought and encountered).

6. Required Training

HazCom training is provided at initial assignment, whenever a new chemical hazard is introduced to the work area, and when this program changes. Records are retained in the LMS.

Training Module	Audience	Frequency
HazCom program overview & employee rights	All employees	Initial + on program change
Reading GHS labels & the 16-section SDS	All chemical-exposed employees	Initial + annual
Locating and accessing SDS on site	All employees	Initial + on system change
Secondary-container labeling	Chemical handlers	Initial + 2-yr refresh
Non-routine task hazard briefing	Affected employees	Before each non-routine task

7. Required Qualifications

- Current completion of HazCom training appropriate to the role, verified in the LMS.
- Ability to locate the SDS system and interpret label elements for chemicals in the assigned area.
- Authorization by the Area Supervisor to handle or work near the chemicals present.

8. Required PPE

HazCom is primarily an information program; PPE is determined by the specific chemical and task per SDS §8 and SOP-EHS-001. When physically applying labels to residue-bearing containers, the following baseline applies.

Task	Eye/Face	Hand	Body
Reviewing SDS / administrative tasks	—	—	—
Labeling clean/sealed containers	Safety glasses	Nitrile gloves	—
Labeling containers with product residue	Goggles	Chemical-resistant gloves	Apron as needed

9. Required Equipment

- Electronic SDS management system accessible from the production floor, with a backup access method during network outage.
- Label printer and GHS-compliant blank label stock (durable, chemical-resistant).
- Current master chemical inventory list linked to the SDS index.
- NFPA 704 placards and workplace-labeling supplies for secondary containers.

10. Hazard Identification

The principal hazard addressed by this SOP is the failure to communicate chemical hazards, which enables downstream physical and health hazards.

- **Uncommunicated hazard** — an unlabeled container or missing SDS leading to unsafe handling or exposure.
- **Misidentification** — wrong label or outdated SDS causing incorrect PPE, storage, or emergency response.
- **Inaccessible information** — SDS not retrievable during an incident, delaying medical or spill response.
- **Untrained exposure** — employees unable to interpret label elements or locate SDS.

11. Risk Assessment

Risk level is derived from the severity × likelihood matrix below. Any residual risk assessed as High or Critical requires additional controls and EHS authorization before the affected work proceeds.

11.1 Risk Rating Matrix

Severity ↓ / Likelihood →	Rare	Unlikely	Possible	Likely	Almost Certain
Catastrophic	Medium	High	High	Critical	Critical
Major	Medium	Medium	High	High	Critical
Moderate	Low	Medium	Medium	High	High
Minor	Low	Low	Medium	Medium	High
Negligible	Low	Low	Low	Low	Medium

11.2 Task Risk Register

Hazard	Severity	Likelihood	Risk Level	Control Measures
Unlabeled secondary container used by another employee	Major	Possible	High	Mandatory workplace labeling, supervisor checks, label stations at points of use.
SDS not accessible during exposure/spill	Major	Unlikely	Medium	Electronic + backup SDS access, emergency SDS retrieval drill, phone-a-service line.
New chemical enters site without SDS	Moderate	Possible	Medium	Procurement gate: no SDS, no receipt; receiving verification (SOP-EHS-001).
Employee cannot interpret label/SDS	Moderate	Possible	Medium	Initial + annual training, quick-reference pictogram guide (Appendix A).
Contractor unaware of area hazards	Major	Unlikely	Medium	Contractor hazard exchange at sign-in; two-way SDS sharing.

12. Detailed Step-by-Step Procedure

12.1 Pre-Job Preparation (Program Maintenance)

1. Maintain a written HazCom program describing labeling, SDS, training, and this SOP; make it available to employees on request.
2. Maintain a master chemical inventory identifying each hazardous chemical by the name used on its label and SDS.
3. Confirm the SDS index cross-references every chemical on the inventory.

NOTE

A chemical must not enter service until its SDS is on file and it is added to the inventory. Procurement enforces this at the point of order and receiving verifies it on delivery.

12.2 SDS Management

1. Obtain the manufacturer/importer SDS with each new chemical; request a current SDS if one is missing or outdated.
2. File the SDS in the electronic system and link it to the inventory record; retain superseded versions per the records schedule.
3. Verify SDS are readily accessible to employees on all shifts, including a backup method usable during a network outage.

12.3 Labeling — Primary Containers

1. Confirm each incoming primary container bears a compliant supplier label: product identifier, signal word, hazard statement(s), pictogram(s), precautionary statement(s), and supplier information.
2. Do not deface, remove, or alter supplier labels. Replace damaged or illegible labels before the container is placed in service.

CAUTION

A container with a missing or unreadable primary label must be quarantined and relabeled from the SDS before use. Never guess a container's contents.

12.4 Labeling — Secondary / Workplace Containers

1. Label every secondary container into which a chemical is transferred, unless it is used immediately by the person who filled it and emptied by end of shift.
2. Apply either the full GHS label elements or a compliant workplace label that conveys the product identifier and general hazard information (e.g., words, pictograms, or an NFPA/HMIS rating supported by training).
3. Verify the label remains legible and firmly affixed; replace as needed.

12.5 Employee Information & Training

1. Provide HazCom training at initial assignment, when a new chemical hazard is introduced to the area, and when this program changes.
2. Cover the location of the written program, inventory, and SDS; how to read labels and SDS; the hazards of area chemicals; and protective measures.
3. Document attendance and comprehension in the LMS (Appendix D acknowledgment).

12.6 Non-Routine Tasks

1. Before a non-routine task (e.g., vessel entry, line breaking, maintenance on chemical systems), brief affected employees on the specific hazards and controls.
2. Coordinate with SOP-EHS-001 and applicable permit procedures; record the briefing.

12.7 Contractor Coordination (Multi-Employer)

1. At sign-in, inform contractors of the hazardous chemicals they may encounter, the labeling system, and how to access SDS.
2. Obtain from contractors the identity and SDS of any hazardous chemicals they bring on site; file these for the duration of the work.

12.8 Verification & Documentation

1. Periodically audit workplace labeling, SDS accessibility, inventory accuracy, and training currency; log findings and corrective actions.
2. Retain training records, inventory, SDS index, and audit records in the DMS per the records schedule.

12.9 Escalation Procedures

1. Missing/illegible label or missing SDS → quarantine the chemical, notify the Area Supervisor, restore compliance before use.
2. New unclassified hazard identified → notify EHS to evaluate and update the inventory, SDS index, and training.
3. Systemic gaps (repeat findings) → EHS escalates to management review and logs per SOP-EHS-013.

13. Emergency Response

WARNING

In any chemical emergency, retrieve the SDS and provide it to responders and medical personnel. The SDS drives correct first aid, fire, and spill actions.

13.1 Immediate Actions

- Identify the chemical from the label; retrieve the SDS (electronic or backup copy).
- Follow the SDS first-aid, fire, and accidental-release measures; alert others and the Area Supervisor.

13.2 Medical Response

- Apply first aid per SDS §4; provide the SDS to medical responders; seek professional care for exposures.

13.3 Fire Response

- Use extinguishing media specified in SDS §5; if beyond incipient stage, activate the alarm and evacuate.

13.4 Spill Response

- Follow SDS §6 and SOP-EHS-003; contain only within training and available equipment.

13.5 Information Access During Emergency

- If the electronic system is unavailable, use the backup SDS access method; a 24-hour emergency contact on the SDS may be called for guidance.

13.6 Notification Chain

Step	Notify	Contact / Method	When
1	Area Supervisor	Radio / floor phone	Immediately on exposure or release
2	EHS On-Call	Ext. 2400 / site radio	Within 5 minutes
3	Emergency Response Team	Emergency line 2222	Release/fire beyond incipient / injury
4	External Emergency Services	911	Life threat, fire, or major release
5	EHS Manager	Per call tree	Within 30 minutes; recordable events

14. Required Documentation

Record	Purpose	Owner	Retention
Written HazCom Program	Describes the site program	EHS Manager	Current + superseded
Master Chemical Inventory (App. E)	Lists all hazardous chemicals on site	Chemical Store Lead	Current + 3 years
SDS Index / Library	Access to hazard information	EHS	30 years (exposure record)
Training Records (LMS)	Evidence of employee training	Training Coordinator	Duration + 3 years
Secondary-Container Label Template (App. B)	Standardizes workplace labels	EHS	Current
Contractor Hazard Exchange	Multi-employer information sharing	Contractor Coordinator	Duration of work + 3 yrs

15. References (Public Standards)

- OSHA 29 CFR 1910.1200 — Hazard Communication, including Appendices C (label elements) and D (SDS).
- UN GHS — Globally Harmonized System of Classification and Labelling of Chemicals.
- NFPA 704 — Identification of the Hazards of Materials for Emergency Response.
- ANSI Z535.4 — Product Safety Signs and Labels.
- ISO 45001:2018; ISO 14001:2015 — Management-system standards.

16. Appendices

Appendix A — GHS Pictogram Quick Reference

Each GHS pictogram is a black symbol on a white field within a red diamond border. Use this quick reference with the SDS; it does not replace the label.

Pictogram (name)	Represents
Flame	Flammables, pyrophorics, self-heating, emits flammable gas, self-reactives, organic peroxides
Flame over circle	Oxidizers
Exploding bomb	Explosives, self-reactives, organic peroxides

Pictogram (name)	Represents
Gas cylinder	Gases under pressure
Corrosion	Skin corrosion/burns, eye damage, corrosive to metals
Skull and crossbones	Acute toxicity (fatal or toxic)
Health hazard	Carcinogen, respiratory sensitizer, reproductive/organ toxicity, mutagenicity, aspiration
Exclamation mark	Irritant, skin sensitizer, acute toxicity (harmful), narcotic effects
Environment	Aquatic toxicity (non-mandatory under OSHA)

Appendix B — Secondary Container Label Template

Label Field	Entry
Product identifier (name)	
Signal word (Danger / Warning)	
Pictogram(s)	☐ Flame ☐ Corrosion ☐ Skull ☐ Health ☐ Exclamation ☐ Oxidizer ☐ Gas ☐ Bomb ☐ Environment
Hazard statement(s)	
Precautionary statement(s)	
Date filled / filled by	

Appendix C — Chemical Labeling Decision Flow

START → Is the container the manufacturer's original with an intact GHS label? — YES → No action; use as labeled.

NO → Is the chemical used immediately by the person who filled it and emptied by end of shift? — YES → Workplace label not required (portable container exception).

NO → Apply a compliant secondary/workplace label: product identifier + hazard information (full GHS elements or trained NFPA/HMIS + words/pictograms).

Is an SDS on file for this chemical? — NO → Do not use; obtain SDS and add to inventory.

YES → Container labeled + SDS on file → Ready for use → END.

Appendix D — HazCom Training Acknowledgment

Field	Entry
Employee name	
Department / Area	
Training date	
Modules completed	☐ Program overview ☐ Labels & SDS ☐ SDS access ☐ Secondary labeling
Chemicals specific to my area reviewed	☐ Yes
Employee signature / date	
Trainer signature / date	

Appendix E — Master Chemical Inventory Template

#	Product (as labeled)	Manufacturer	SDS on file	Location / Area	Primary hazard class
1			☐		

#	Product (as labeled)	Manufacturer	SDS on file	Location / Area	Primary hazard class
2			☐		
3			☐		
4			☐		
5			☐		